Chordal and Conference Cubics: Reconstruction and a Residual C_2-Torsor

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Abstract

Different lossy invariants of the same source need not have the same geometry. We ask when two such cubic shadows nevertheless reconstruct one another and their common six-axis data. Let Ω be the six Sylow-5 subgroups of A_5, and let V be the five-dimensional augmentation module of $\mathbf{F}_{11}^{\Omega}$, with its standard quadratic form. The A_5-invariant cubic pencil in $\mathbf{P}(V)$ contains a conference cubic with six isolated nodes and two chordal cubics, each singular along a rational normal quartic. Over $\overline{\mathbf{F}}_{11}$, the conference cubic is not projectively isomorphic to either chordal cubic. We prove that they nevertheless encode the same marked carrier. The singular quartic recovers the constant double cover $A_5/C_5 \to \Omega = A_5/D_{10}$. If either chordal line L is selected, where L is a one-dimensional subspace of the invariant pencil, the outer involution q of that pencil gives mutually inverse reconstruction functors between a normalized chordal generator $h \in L$ and an oriented conference generator c, for neutral scalar extensions over every field extension of $\mathbf{F}_{11}$. Forgetting L quotients by the free involution $(L, h, c) \mapsto (qL, -qh, c)$, which fixes c; this is distinct from the conference-orientation torsor below.

The conference locus also has an intrinsic six-point recognition theorem. Let S represent a two-graph Δ on a six-set X, with $S_{xx} = 0$ and $S_{xy} = S_{yx} \in \{\pm 1\}$. Put $\sigma(xyz) = S_{xy}S_{yz}S_{zx}$ and $m(xy) = \sum_{z \notin \{x,y\}} \sigma(xyz)$. If $A(\Delta)$ is the family of four-subsets on which all four triple signs agree, then

$$16|A(\Delta)| = \sum_{\{x,y\} \in \binom{X}{2}} m(xy)^2.$$

Consequently $A(\Delta) = \varnothing$ exactly when $S^2 = 5I$, the conference condition.

The recovered six-set also controls an integral normalization. For every symmetric conference matrix B of order $n \equiv 2 \pmod 4$, normalized so that the off-diagonal entries in its first row are $+1$, $D_n^{\vee} = \mathbf{Z}^n + \mathbf{Z}\mathbf{1}/2$ is the least $\varphi = (I + B)/2$-stable lattice containing $\mathbf{Z}^n$. Moreover $\varphi^2 - \varphi = (n-2)I/4$, so the algebra $\mathbf{F}_2[\bar{\varphi}]$ induced on $D_n^{\vee}/2D_n^{\vee}$ is $\mathbf{F}_4$ for $n \equiv 6 \pmod 8$ and $\mathbf{F}_2 \times \mathbf{F}_2$ for $n \equiv 2 \pmod 8$. At $n = 6$, the residue is the unique nonsplit extension, up to isomorphism of $\mathbf{F}_4A_5$-modules, of a trivial line by the natural module H, with $\mathrm{End}_{A_5}(H) = \mathbf{F}_4$. The map $[B] \mapsto \bar{\varphi}|_H$ identifies $\{[B], [-B]\}$ with the two primitive scalars $\{\omega, \omega^2\} \subset \mathbf{F}_4$; conference reversal corresponds to Frobenius $z \mapsto z^2$. This is an equivariant identification of principal C_2-torsors.

Keywords. chordal cubic; conference matrix; reconstruction; root lattice; modular representation; Frobenius descent.

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1 Introduction and statement

Different lossy invariants of the same source need not look alike. The inverse question is whether they nevertheless retain equivalent information. On the irreducible five-dimensional metric A_5-module, the invariant cubic pencil contains geometrically different members: a conference line and two chordal lines, each chordal cubic singular along a rational normal quartic. We prove that, after one chordal line is retained, either shadow reconstructs the other and their common six-axis carrier. Forgetting that line leaves exactly a C_2-ambiguity. The result classifies information loss between inequivalent cubic shadows; it does not identify the cubics. The classification itself is formulated on the fixed metric carrier.

The explanation has four steps. First, the singular quartic of a chordal cubic contains a distinguished split A_5/C_5-orbit. Sending each point to its stabilizer produces the canonical quotient

$$A_5/C_5 \longrightarrow A_5/D_{10},$$

and hence recovers the six-axis carrier. Second, the nontrivial outer permutation of this six-set acts linearly on the invariant pencil. Its difference from the identity carries either selected chordal line isomorphically onto the conference line. Third, retaining the standard quadratic form on the augmentation removes the scalar inertia that a cubic alone would leave after field extension. The resulting categories are therefore honest metric groupoids, and their exact forgetful fibres can be computed. Fourth, the same recovered six-set supports an integral root–weight normalization. In order six, its residue is the natural $\mathbf{F}_4A_5$-heart, and outer reversal acts on it by Frobenius.

The conference object has a second intrinsic description on the recovered carrier. If $m(xy)$ is the signed defect of a pair, the number of aligned four-sets is $16^{-1}\sum m(xy)^2$. Hence the alignment-invisible six-point two-graphs are exactly the conference ones. This recognition is weaker than the recovered A_5-marking—there are twelve labeled conference switching classes—but the marking selects the invariant opposite pair used by the outer-difference construction.

The exceptional six-point dictionaries and Joubert–Segre constructions are classical; we use Howard–Millson–Snowden–Vakil [7, §§1.1–1.6 and 2.1–2.4]. The invariant pencil and its chordal members are due, in the form needed here, to Pinardin–Zhang [10, §6.2]. Their statement is over characteristic zero; Lemma 4.2 proves that the same two-point chordal scheme is reduced and exhaustive in characteristic eleven. The unmarked chordal geometry has the larger automorphism group PGL_2 [4, §5]. Conference switching follows Goethals–Seidel [5, §§1–3], and the order-six switching class is classical [1]. Gillespie studies coherent and incoherent four-set counts for regular two-graphs [6, §4.1], while the fourth-power equality criterion for conference Seidel matrices appears in Iranmanesh–Askari Farsangi [9, Theorem 2.4]. Lemma 3.4 gives the direct order-six defect count used here. Our claims concern the marked compatibility, normalization, exact information loss, and the root–weight residue obtained when these ingredients are composed. We make no priority claim beyond this attribution boundary.

Put $k = \mathbf{F}_{11}$, $G = A_5$, and let

$$\Omega_0 = \mathrm{Syl}_5(G), \qquad A_0 = \mathrm{Aug}(k^{\Omega_0}), \qquad Q_0(x) = \sum_{\omega\in\Omega_0} x_\omega^2.$$

Thus $|\Omega_0| = 6$, A_0 has dimension five, and in the basis $e_i - e_6$ the Gram matrix of Q_0 is $I + J$, of determinant 6. Write

$$\Pi = \mathrm{Sym}^3(A_0^*)^G.$$

The standard order-six conference class determines an unordered pair of opposite actual cubics $\{\pm c_B\} \subset \Pi$. A literal outer permutation of Ω_0 induces a linear involution q_Π on Π; its definition is independent of the chosen representative of the outer coset because G acts trivially on Π.

Definition 1.1 (Metric chordal shadow)**.** Let K/k be a field extension. A metric chordal shadow over K is the neutral scalar extension of (A_0, Q_0), together with a nonzero actual cubic $h \in \Pi_K$ whose saturated Jacobian scheme is a rational normal quartic. If

$$(q_\Pi - 1)h = \alpha c_B,$$

for one of the coefficient-normalized conference generators, the shadow is *normalized* when $\alpha^2 = 8^2$. Morphisms are G-equivariant Q_0-isometries with a specified normalizer relabeling, as defined in Definition 5.1; equivariance may be twisted only by the automorphism induced by that relabeling.

The word *neutral* is deliberate. We classify scalar extensions of this fixed quadratic carrier, not arbitrary twisted K-forms. The quadratic form is also essential: over a field containing μ_3, a scalar ζI with $\zeta^3 = 1$ fixes every cubic; preservation of Q_0 adds $\zeta^2 = 1$, and hence forces $\zeta = 1$.

Our main result is most compactly stated as follows. The groupoids named in the theorem are defined independently in Section 5.

Theorem 1.2 (Intrinsic companion classification)**.** *For every extension K/k, neutral scalar extension gives natural equivalences*

$$\mathscr{C}^L_{\mathrm{conf}}(K) \simeq \mathscr{G}_{\mathrm{met}}(K) \simeq \mathscr{C}_{\mathrm{ch}}(K).$$

Here the first groupoid consists of oriented order-six conference packages with a selected compatible chordal line, the second of the derived expanded metric carriers, and the third of normalized metric chordal shadows. Under these equivalences:

(i) *the singular quartic recovers a split degree-twelve scheme Z_5 and a canonical stabilizer quotient $Z_5 \to \Omega_0$;*

(ii) *the oriented conference generator is*

$$c = 8^{-1}(q_\Pi - 1)h;$$

(iii) *the selected-line functors return all retained source data of Papers I–III, but no source-local chart or global cover not included in the bridge datum;*

(iv) *after the selected chordal line is forgotten,*

$$\mathscr{C}_{\mathrm{conf}}(K) \simeq \mathscr{C}_{\mathrm{ch}}(K)/\langle uq\rangle,$$

where $u = -I$ and $uq : (L, h, c) \mapsto (qL, -qh, c)$. Thus the unselected map is a residual C_2-torsor, not an equivalence.

Theorem 1.3 (Normalization–residue theorem)**.** *The recovered six-set canonically distinguishes the rank-five augmentation lattice from the rank-six D-type lattice, while identifying their common binary heart. For every normalized symmetric conference matrix B of order $n \equiv 2 \pmod 4$, the operator*

$$\varphi_B = \frac{I + B}{2}$$

minimally stabilizes $D_n^\vee$, and

$$\varphi_B^2 - \varphi_B = \frac{n-2}{4} I.$$

Modulo two its coefficient algebra is $\mathbf{F}_4$ *when* $n \equiv 6 \pmod 8$ *and* $\mathbf{F}_2 \times \mathbf{F}_2$ *when* $n \equiv 2 \pmod 8$. *At* $n = 6$, *writing* $M = D_6^\vee / 2D_6^\vee$, *there is a canonical nonsplit sequence*

$$0 \longrightarrow H \longrightarrow M \longrightarrow \ell \longrightarrow 0,$$

where H *is a natural two-dimensional* $\mathbf{F}_4 A_5$*-module and* $\ell = M/H$ *is the trivial* $\mathbf{F}_4$*-line. After choosing a trivialization* $\ell \cong \mathbf{F}_4$,

$$\mathrm{Ext}^1_{\mathbf{F}_4 A_5}(\mathbf{F}_4, H) \cong \mathbf{F}_4.$$

On the endomorphism field

$$\mathrm{End}_{A_5}(H) = \mathbf{F}_4,$$

golden reversal $B \mapsto -B$ *and the outer normalizer both restrict to Frobenius. Relative to the recovered six-set, the golden-orientation torsor and the exotic Frobenius torsor are therefore the same principal* C_2*-torsor.*

Fix now the Paper-II H_3 matching

$$M = \{01, 25, 37, 49, 68, 10\,11\}$$

on the twelve points of $\mathbf{P}^1(\mathbf{F}_{11})$. Its stabilizer in $\mathrm{PSL}_2(11)$ is a group $G \cong A_5$, acting transitively on the six pairs of M. Write X_M for that six-set and

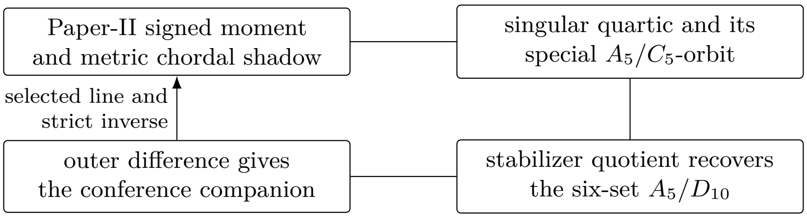
$$A_M = \{(u_x)_{x \in X_M} \in \mathbf{F}_{11}^{X_M} : \textstyle\sum_x u_x = 0\}$$

for its augmentation module. Paper II constructs a ten-dimensional quotient W, an intrinsic pair of matching sheets, and their first nonzero signed moment $\mu_3 \in \mathrm{Sym}^3(W)$ [11]. Since the action on X_M is transitive, a pair stabilizer has order ten; the order-ten subgroups of A_5 are the self-normalizing groups D_{10}. Thus $X_M \cong A_5/D_{10}$.

Let W_5 be the five-isotypic summand of $W|_G$. The invariant self-duality of W_5 turns the projection of μ_3 into a projective cubic line $[H_M] \subset \mathbf{P}(\mathrm{Sym}^3(W_5^*))$.

The retained Paper-II datum is the actual projected generator together with its normalized multiplicity-one bridge to (A_0, Q_0). The target metric is the fixed Q_0; no isomorphism class of arbitrary quadratic forms is being classified. Section 2 proves that the image is one of the normalized metric chordal shadows in Theorem 1.2.

Proof and reading map.



Section 2 gives the one Paper-II placement calculation. Sections 3 and 4 prove the special orbit and outer-difference bridge. Section 5 upgrades the round trip to an intrinsic classification and computes its marking fibres. Section 6 records the source-specific returns. Sections 7–9 prove the integral normalization–residue theorem. Section 10 compares its C_2-rigidification with Paper IV's separate C_3-rigidification.

Scope and exact information loss. The classification is intrinsic after fixing the literal quadratic augmentation carrier and concerns its neutral scalar extensions. It does not classify arbitrary twisted forms, identify the chordal and conference cubics, or manufacture source-local charts not retained in the bridge datum. These boundaries record the information actually present in the shadows. Likewise, the integral theorem does not identify the two integral lattices; it identifies their common binary heart and the residual marking carried there.

2 The metric chordal normal form

The restriction of Paper II's quotient to G has character

class	1	2	3	$5A$	$5B$
class size	1	15	20	12	12
χ_W	10	2	1	0	0

and hence

$$W|_G \cong \mathbf{1} \oplus V_4 \oplus V_5.$$

The quotient and signed moment are imported from Paper II's stable results *The* $3, 6, 10$ *quotient ranks* and *Balanced sheets and cubic-first orientation* [11]; the displayed restriction is the character calculation used here. The field $\mathbf{F}_{11}$ contains the fifth roots of unity and $\sqrt{5} = 4$, and $11 \nmid 60$, so it is a splitting field for this calculation and the summands are absolutely irreducible. Because $11 \nmid 60$, the central character idempotent

$$e_5 = \frac{5}{60} \sum_{g \in G} \chi_5(g) g$$

is defined over $\mathbf{F}_{11}$, is idempotent, and has rank five. The degree-six permutation character is $\mathbf{1} + \chi_5$, so $A_M \cong V_5$. Schur's lemma gives

$$\dim_{\mathbf{F}_{11}} \operatorname{Hom}_G(A_M, e_5 W) = 1.$$

Proposition 2.1 (Paper-II placement)**.** *After the outer-twisted multiplicity-one identification with* A_0*, the chosen-sheet projection of Paper II's signed moment is a metric chordal shadow whose actual cubic is normalized by its first nonzero coefficient. Its projective equation in standard binary-quartic coordinates is*

$$H = z_0 z_2 z_4 + 2 z_1 z_2 z_3 - z_0 z_3^2 - z_1^2 z_4 - z_2^3 = \det \begin{pmatrix} z_0 & z_1 & z_2 \\ z_1 & z_2 & z_3 \\ z_2 & z_3 & z_4 \end{pmatrix}.$$

Its sheet reversal is $-H$*. In the normalization used below, the Paper-II pivot is* 3*, and the outer-difference coefficient is* 8*.*

Proof. The representation-theoretic reduction is the calculation above. Both A_M and $e_5 W$ are copies of the absolutely irreducible module V_5, so their bridge is unique up to scalar. There are two comparisons with the binary-quartic model: the direct identification and its outer twist. The geometric discriminator is whether the Veronese quartic becomes the singular curve. The direct comparison fails this test and the outer-twisted comparison passes it.

For completeness, we print the normalization leaf. Choose the axis basis $e_i - e_6$ of A_M and Paper II's reduced-pivot basis of W_5. The bridge line is generated by

$$T = \begin{pmatrix} 2 & 9 & 7 & 6 & 4 \\ 0 & 9 & 10 & 10 & 6 \\ 5 & 1 & 9 & 3 & 0 \\ 0 & 3 & 10 & 1 & 6 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The defining generator relations of G verify $\rho_{W_5}(g)T = T\rho_{A_M}(g)$, and $\det T \neq 0$. Put $B_0 = I + J$, the Gram matrix of Q_0. If $\nu \in \mathrm{Sym}^3(W_5)$ is the chosen-sheet projection of μ_3, its axis cubic is

$$\widetilde{h}_M = \mathrm{Pol}\Big(\mathrm{Sym}^3(B_0 T^{-1})\nu\Big).$$

Here is the complete finite calculation. Order the cubic monomials as

$$\mathcal{M} = (x_i x_j x_k)_{0 \le i \le j \le k \le 4}.$$

In that order the coefficient vector of $\widetilde{h}_M$ is

$$\begin{gathered}(0, 3, 3, 5, 5, 3, 6, 10, 6, 3, 6, 10, 5, 10, 5, 0, 5, 5, \\ 3, 5, 10, 10, 5, 6, 3, 0, 3, 5, 3, 6, 5, 0, 3, 3, 0).\end{gathered}$$

Thus its first nonzero coefficient, that of $x_0^2 x_1$, is 3, so set $h_M = 3^{-1}\widetilde{h}_M$. The displayed vector is the result of the three operations in the preceding formula—the central projection, the intertwiner, and metric duality—and is an independently checkable record of the only Paper-II coordinate transport used here. The scalar is fixed before the sheet is reversed. Hence the other sheet is $-h_M$, rather than a separately projectivized generator.

After the nontrivial element of $\mathrm{Out}(A_5)$, the unique binary-quartic projectivity is

$$U = \begin{pmatrix} 0 & 0 & 3 & 5 & 3 \\ 4 & 1 & 1 & 10 & 4 \\ 3 & 6 & 3 & 0 & 0 \\ 5 & 1 & 6 & 10 & 5 \\ 9 & 10 & 10 & 7 & 1 \end{pmatrix}.$$

With the pullback convention $U(h)(z) = h(Uz)$, substitution in the displayed coefficient vector gives the polynomial identity

$$U(h_M) = 8H.$$

Indeed, the coefficients of $z_0 z_2 z_4, z_0 z_3^2, z_1^2 z_4, z_2^3, z_1 z_2 z_3$ are respectively $8, -8, -8, -8, 16$, and every other coefficient is zero. Thus the equality is visible term by term, rather than inferred from a singular-point census. The conference cubic has six isolated nodes by Paper I and therefore spans a different line. This proves the placement without an exhaustive cubic comparison. □

The displayed matrices only fix the bridge between the frozen Paper-II basis and the intrinsic carrier. Every subsequent argument uses the recovered metric shadow, its special orbit, and its normalizer. The exact replay named in Section 11 checks this normalization leaf independently, but no later theorem depends on a coefficient census.

3 The singular quartic recovers the axes

Lemma 3.1 (Singular locus of the Hankel determinant)**.** *Over $\mathbf{F}_{11}$, the singular scheme of $H = 0$ is the rational normal quartic R.*

Proof. The five partial derivatives are

$$\begin{aligned} z_2z_4 - z_3^2, &\qquad 2(z_2z_3 - z_1z_4), \\ z_0z_4 + 2z_1z_3 - 3z_2^2, &\qquad 2(z_1z_2 - z_0z_3), \\ z_0z_2 - z_1^2. & \end{aligned}$$

Let I_R be the ideal of the six 2-by-2 minors of

$$\begin{pmatrix} z_0 & z_1 & z_2 & z_3 \\ z_1 & z_2 & z_3 & z_4 \end{pmatrix}.$$

These minors define the rational normal quartic scheme. Four are scalar multiples of the first, second, fourth, and fifth displayed derivatives. Write the other two as

$$A = z_0z_4 - z_1z_3, \qquad B = z_1z_3 - z_2^2.$$

The middle derivative is $A + 3B$, so the Jacobian ideal J is contained in I_R. Conversely, on $D(z_2)$, the identity

$$z_2A = z_0(z_2z_4 - z_3^2) + z_3(z_0z_3 - z_1z_2)$$

puts A, and then B, in J_{z_2}. On $D(z_0)$,

$$z_0B = z_1(z_0z_3 - z_1z_2) - z_2(z_0z_2 - z_1^2)$$

does the same; the symmetric identity handles $D(z_4)$. These three opens cover the projective Jacobian scheme, since $z_0 = z_2 = z_4 = 0$ and the outer two derivatives force $z_1 = z_3 = 0$. Hence the saturated Jacobian ideal equals I_R, proving the scheme-theoretic assertion. □

Proposition 3.2 (The special orbit and its stabilizer quotient)**.** *Let R be the singular quartic of a metric chordal shadow. The scheme*

$$Z_5 = \coprod_{P \in \mathrm{Syl}_5(G)} R^P$$

is split finite étale of degree twelve. Every geometric point has exact stabilizer C_5, and the stabilizer map is the canonical double cover

$$Z_5 \longrightarrow \Omega_0.$$

For the Paper-II shadow this quotient is the original six-set X_M.

Proof. The rational normal quartic spans $\mathbf{P}(A_0)$, so an element acting trivially on R acts projectively trivially on A_0. The augmentation action is faithful, and therefore G acts faithfully on $R \cong \mathbf{P}^1$, hence through $\mathrm{PGL}_2(11)$. Its composite with the determinant-class quotient $\mathrm{PGL}_2(11)/\mathrm{PSL}_2(11) \cong C_2$ is trivial because $G \cong A_5$ is simple. Thus G lies in $\mathrm{PSL}_2(11)$. For $x \in R(\mathbf{F}_{11})$, its stabilizer lies in the order-55 Borel subgroup of $\mathrm{PSL}_2(11)$, hence has order dividing $\gcd(60, 55) = 5$. A Sylow 5-subgroup is split and fixes two points. Distinct Sylow 5-subgroups cannot share a fixed point, since the point stabilizer has order at most five. The six Sylow subgroups therefore supply $6 \cdot 2 = 12$ distinct points, exhausting

$R(\mathbf{F}_{11})$. Every point stabilizer is C_5, the action is transitive, and the six fixed-point pairs form

$$G/N_G(C_5) \cong A_5/D_{10}.$$

The stabilizer of each pair of the original matching is also D_{10}, and this subgroup is self-normalizing in G. Hence it fixes a unique matching pair in the transitive G/D_{10} action. Sending a singular point to the matching pair fixed by the normalizer of its stabilizer therefore gives the intrinsic map

$$A_5/C_5 \longrightarrow A_5/D_{10}, \qquad gC_5 \longmapsto gN_G(C_5),$$

with fibres of size two. This proves the assertion over k.

The fixed points of a generator of C_5 are its two distinct eigenlines, already defined over k because $\mu_5 \subset k^\times$. Thus the six pairs are reduced, disjoint, and constant. The description commutes with every scalar extension K/k, proving the scheme statement. $\square$

Lemma 3.3 (The intrinsic conference pair)**.** *The recovered G-set $\Omega_0 \cong A_5/D_{10}$ determines a unique unordered pair $\{[B], [-B]\}$ of G-invariant order-six conference switching classes. Coefficient normalization gives an unordered pair of actual triangle cubics $\{\pm c_B\} \subset \Pi$.*

Proof. Normalize a symmetric conference signing by switching so that its first row is positive. Orthogonality with that row forces every other row to have two positive and two negative entries away from the first coordinate and its diagonal. The negative entries therefore form a simple two-regular graph on five vertices, hence a pentagon. All pentagons are relabelled copies of one another; complementing the pentagon gives the opposite switching class. The natural six-point A_5-action preserves this class, and its two orientations give opposite triangle cubics on the augmentation. More precisely, the automorphism group of an oriented order-six conference class is the transitive A_5. Identify it with the recovered $G \leq S(\widehat{X}_M)$. Two such identifications differ through

$$N_{S_6}(G)/G \cong C_2,$$

and the nontrivial coset exchanges the class with its opposite. Hence the unordered opposite pair, and therefore its projective triangle-cubic line, is intrinsic to the fixed recovered G-set. This classical uniqueness and pentagon representative also appear in Bussemaker–Mathon–Seidel [1, pp. 12, 21, and Table 9 on p. 83]; our switching convention follows Goethals–Seidel [5, §§1–3]. $\square$

Lemma 3.4 (Six-point alignment recognition)**.** *Let Δ be a two-graph on a six-set Ω, represented by a Seidel matrix S, so $S_{xx} = 0$ and $S_{xy} = S_{yx} \in \{\pm 1\}$ for $x \neq y$. Put*

$$\sigma(xyz) = S_{xy}S_{yz}S_{zx}, \qquad m(xy) = \sum_{z \notin \{x,y\}} \sigma(xyz) = S_{xy}(S^2)_{xy}.$$

Call $Q \in \binom{\Omega}{4}$ aligned when σ is constant on $\binom{Q}{3}$, and write $A(\Delta)$ for the family of aligned four-sets. Then

$$16\,\#\{Q \in \tbinom{\Omega}{4} : Q \text{ is aligned}\} = \sum_{\{x,y\} \in \binom{\Omega}{2}} m(xy)^2.$$

Consequently Δ has no aligned four-set if and only if $S^2 = 5I$. Relative to the recovered A_5-action, the unordered pair of Lemma 3.3 is therefore the pair of A_5-fixed two-graphs in the empty-alignment locus.

Proof. Fix $t = \{x, y, z\}$ and $w \notin t$. Set

$$\alpha = \sigma(xyw)\sigma(t), \qquad \beta = \sigma(xzw)\sigma(t), \qquad \gamma = \sigma(yzw)\sigma(t).$$

Each pair in $t \cup \{w\}$ occurs in two of its four triples, so $\alpha\beta\gamma = 1$. The indicator that $t \cup \{w\}$ is aligned is therefore

$$\frac{(1+\alpha)(1+\beta)(1+\gamma)}{8} = \frac{1+\alpha+\beta+\gamma}{4}.$$

Summing over the three choices of w gives

$$\lambda_3(t) = \frac{\sigma(t)}{4} \sum_{p \in \binom{t}{2}} m(p),$$

where $\lambda_3(t)$ counts aligned four-sets through t. Every aligned four-set contains four triples, so a second summation yields

$$4|A(\Delta)| = \frac{1}{4} \sum_t \sigma(t) \sum_{p \in \binom{t}{2}} m(p) = \frac{1}{4} \sum_p m(p)^2.$$

This is the claimed identity. Finally $m(xy) = S_{xy}(S^2)_{xy}$, while every diagonal entry of S^2 is five. Thus all defects vanish exactly when $S^2 = 5I$. Diagonal switching fixes every triple sign. Global complementation $S \mapsto -S$ negates σ and m, so both the aligned family and the square identity are intrinsic to the two-graph up to complement. $\square$

Paper III's theorem *Aligned-design faithfulness* proves that aligned four-sets determine a two-graph up to complement from seven points onward, with seven sharp [13]. The lemma identifies the empty-alignment fibre at that sharp six-point boundary: it is precisely the unmarked conference locus. The recovered A_5-action then selects the opposite pair needed here.

Paper I's corollary *Nodes, symmetry, and integral commutant* proves that this triangle cubic has exactly six ordinary nodes in projective-frame position [12]. Since H_M is singular along a curve, the conference and chordal lines are distinct. The point is causal: the conference pair was constructed from the recovered six-set, not inserted among the recognition axioms.

4 Outer difference and the oriented companion

The ordinary character of the augmentation module is

$$\chi_5 = (5, 1, -1, 0, 0)$$

on the classes $1, 2A, 3A, 5A, 5B$. The symmetric-cube formula gives

$$\chi_{\mathrm{Sym}^3 A_0^*} = (35, 3, 2, 0, 0),$$

and therefore

$$\dim \Pi = \frac{35 + 15 \cdot 3 + 20 \cdot 2}{60} = 2.$$

Maschke's theorem and the Reynolds idempotent show that this calculation commutes with every extension K/k.

The construction of the outer operator is linear, not merely projective. Let τ be an involutive permutation in the nontrivial coset of $N_{S(\Omega_0)}(G)/G$, and let P_τ act on the literal augmentation A_0. If an abstract copy A of the five-dimensional module is in use and $T : A \to A_0$ is any nonzero intertwiner, put

$$q = T^{-1} P_\tau T.$$

Changing the scalar of T changes nothing. Changing τ by an element of G changes nothing on Π. Thus the induced $q_\Pi \in \mathrm{GL}(\Pi)$ is canonical relative to the recovered six-set.

Proposition 4.1 (Normalized outer action)**.** *Let c_B be either coefficient-normalized conference generator and let h be the coefficient-normalized Paper-II chordal generator of Proposition 2.1. In the ordered basis (c_B, h),*

$$[q_\Pi] = \begin{pmatrix} -1 & 8 \\ 0 & 1 \end{pmatrix}.$$

Consequently the conference line is $\Pi^- = \ker(q_\Pi + 1)$, the two chordal lines are exchanged, and

$$(q_\Pi - 1)h = 8c_B.$$

In particular, h is normalized in the sense of Definition 1.1.

Proof. Take the literal odd permutation $(3\ 4\ 5\ 6)$ of the recovered six-set. In the basis $e_i - e_6$, its augmentation matrix is

$$Q = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & -1 & -1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Reversing the conference class changes every triangle sign, hence sends c_B to $-c_B$. Proposition 2.1 identifies the Paper-II line with the chordal line and fixes its pivot scalar. Comparing one coefficient now makes the remaining scalar transparent. With the monomial order $\mathcal{M}$ of Proposition 2.1, coefficient normalization gives

$$[x_0^2 x_1]c_B = 1, \qquad [x_0^2 x_1]h = 1, \qquad [x_0^2 x_1]q_\Pi h = 9.$$

Thus $[x_0^2 x_1](q_\Pi h - h) = 8$. Since both sides lie in the one-dimensional anti-invariant line, this comparison proves the displayed formula. The independence of the outer representative was proved above. $\square$

Lemma 4.2 (The two chordal lines)**.** *Over $\overline{k}$, the chordal-member subscheme of $\mathbf{P}(\Pi)$ consists of exactly two reduced points,*

$$[h], \qquad [q_\Pi h].$$

Both are defined over k. Consequently every metric chordal shadow over an extension K/k lies on one of their scalar extensions.

Proof. Let $f \in \Pi_{\overline{k}}$ be chordal and let R_f be its singular rational normal quartic. Since f is G-invariant, R_f is G-stable. Its action on $R_f \cong \mathbf{P}^1$ is faithful: the kernel is normal in the simple group G, and if all of G acted projectively trivially on R_f, it would act projectively trivially on the span $\mathbf{P}(A_0)$.

Thus R_f determines a faithful two-dimensional projective representation of A_5. Because $11 \nmid 120$, the representation theory of the binary icosahedral cover is semisimple and has the same character table as in characteristic zero. Its irreducible degrees are

$$1, 2, 2, 3, 3, 4, 4, 5, 6;$$

the two degree-two characters are faithful and are exchanged by $\mathrm{Out}(A_5)$. For either representation, say V, its fourth symmetric power is the five-dimensional module A_0, and

$$\dim \mathrm{Hom}_G(\mathrm{Sym}^4 V, A_0) = 1.$$

Hence each projective representation yields at most one equivariant Veronese quartic in $\mathbf{P}(A_0)$, and its secant cubic is unique. There are therefore at most two chordal lines. Proposition 2.1 supplies one, and its image under the literal outer permutation supplies the other. They are distinct because $(q_\Pi - 1)h = 8c_B \neq 0$. Both are defined over k.

It remains to exclude an infinitesimal thickening. A first-order chordal deformation carries its scheme-theoretic singular quartic with it. The projective action on that quartic has no infinitesimal deformation because $H^1(G, \mathfrak{pgl}_2) = 0$: averaging is valid since $|G|$ is invertible in k. Once the projective action is fixed, the equivariant Veronese embedding has no projective tangent because the preceding Hom space is a line. Thus both points are reduced. The final assertion follows after base change. $\square$

Corollary 4.3 (Selected-line bridge)**.** *For either chordal line $L \subset \Pi$, the restriction*

$$q_\Pi - 1 : L \longrightarrow \Pi^-$$

is an isomorphism. On the punctured lines it is $K^\times$-equivariant. On normalized generators its forward and inverse maps are

$$h \longmapsto c = 8^{-1}(q_\Pi - 1)h, \qquad c \longmapsto h = ((q_\Pi - 1)|_L)^{-1}(8c).$$

They carry $-h$ to $-c$.

Proof. The assertion on the Paper-II line is Proposition 4.1. By Lemma 4.2, the other chordal line is $q_\Pi L$, and $q_\Pi^2 = 1$, so the restriction there is nonzero as well. Every remaining assertion follows from linearity. $\square$

The line L is not cosmetic. The conference package remembers the unordered pair $\{L, q_\Pi L\}$, but no one member of it. Moreover

$$(q_\Pi - 1)(-q_\Pi h) = (q_\Pi - 1)h.$$

Thus h and $-q_\Pi h$ have the same oriented conference companion. This residual involution is the precise obstruction to an unselected equivalence.

5 Intrinsic groupoids and exact marking fibres

We now define the classification categories without referring to the image of a reconstruction functor.

Put $N = N_{S(\Omega_0)}(G)$. For $\tau \in N$, write $\sigma_\tau(g) = \tau g \tau^{-1}$ and let P_τ be its permutation isometry on the literal augmentation module. A *metric normalizer morphism* is a pair (τ, F) with

$$\tau \in N, \qquad F \in O(A_{0,K}, Q_0), \qquad F\rho(g) = \rho(\sigma_\tau(g))F,$$

whose projectivization sends the six augmentation-frame points according to τ. On cubics we use the covariant convention $F_{\#}f = f \circ F^{-1}$. The morphism must carry every retained line and actual tensor by $F_{\#}$. For a conference class it must also satisfy

$$[P_\tau B P_\tau^{-1}] = [B'],$$

where brackets mean equivalence under diagonal switching $B \sim DBD$, with

$$D \in \{\pm 1\}^{\Omega_0}/\{\pm I\}.$$

Thus switching is an explicit gauge on the conference representative; it is not silently identified with a linear automorphism of the augmentation hyperplane. Composition is $(\tau', F') \circ (\tau, F) = (\tau'\tau, F'F)$.

Definition 5.1 (The three companion groupoids)**.** For an extension K/k, let:

(a) $\mathscr{C}_{\mathrm{ch}}^{\times}(K)$ be the groupoid of neutral metric chordal shadows with arbitrary nonzero actual generator, and let $\mathscr{C}_{\mathrm{ch}}(K)$ be its full normalized subgroupoid $\alpha^2 = 8^2$, with metric normalizer morphisms carrying h to h';

(b) $\mathscr{G}_{\mathrm{met}}(K)$ be the groupoid of tuples derived from such a normalized shadow:

$$(A_0, Q_0, \Pi, R_h, Z_5, \Omega_0, q_\Pi, L, h, c, [B]);$$

every derived entry $R_h, Z_5, \Omega_0, q_\Pi, L, c, [B]$ is required to be the construction of Sections 3 and 4, not an independent recognition axiom, and morphisms are metric normalizer morphisms carrying every displayed entry;

(c) $\mathscr{C}_{\mathrm{conf}}^{L}(K)$ be the groupoid of oriented order-six conference packages $(\Omega_0, [B], c)$, where $c = c_B$ is the coefficient-normalized actual triangle cubic attached to the oriented switching class $[B]$; replacing $[B]$ by $[-B]$ replaces c by $-c$. The package also includes a selected member L of the two-point chordal scheme of Lemma 4.2. Its morphisms are the metric normalizer morphisms above that carry $[B], c, L$ to $[B'], c', L'$.

Let $\mathscr{C}_{\mathrm{conf}}(K)$ be obtained by forgetting L.

This definition does not assume the conclusions that make the construction work. The chordal scheme, the special orbit, the conference pair, and the scalar 8 were proved before the groupoids were introduced. In particular, the conference-side object does not assume that its inverse is a chordal shadow; that fact follows from the selected line and Corollary 4.3.

Proposition 5.2 (Strict rigidity)**.** *The functors*

$$\mathscr{C}_{\mathrm{ch}}(K) \longrightarrow \mathscr{G}_{\mathrm{met}}(K) \longleftarrow \mathscr{C}_{\mathrm{conf}}^{L}(K)$$

are fully faithful and essentially surjective.

Proof. Starting with a chordal shadow, Lemma 3.1 and Proposition 3.2 recover R_h, Z_5, and Ω_0. Lemma 3.3 recovers the conference pair, and the literal normalizer recovers q_Π. Proposition 4.1 then identifies the normalized orientation

$$c = 8^{-1}(q_\Pi - 1)h.$$

This proves essential surjectivity from the chordal side. Starting from the selected-line conference package, Corollary 4.3 recovers h, after which the same construction returns every derived entry. This proves essential surjectivity from the conference side.

For full faithfulness, fix a normalizer label τ. A projectivity carrying the recovered six-point frame according to τ is unique, so any two linear lifts differ by a scalar λ. The metric and the actual cubic impose $\lambda^2 = \lambda^3 = 1$, hence $\lambda = 1$; this is where the metric removes the hidden μ_3-inertia. Thus the pair (τ, F) is determined on either side. The singular scheme is functorial under F, the stabilizer map is functorial under σ_τ, and the conference pair and q_Π are defined from the literal normalizer action. Consequently preservation of h forces preservation of every derived entry and of $c = 8^{-1}(q_\Pi - 1)h$.

Conversely, a conference-side metric normalizer morphism carries the selected line and c, hence carries $((q_\Pi - 1)|_L)^{-1}(8c)$ to the corresponding inverse on the target. The switching condition in the definition is exactly what is needed to transport $[B]$, and no diagonal switch acts as an extra scalar on the augmentation carrier. Therefore the two Hom maps are bijective. □

Proposition 5.2 proves the selected-line equivalences in Theorem 1.2. The unselected statement uses the action groupoid for the autoequivalence

$$uq : (L, h, c, [B]) \longmapsto (qL, -qh, c, [B]),$$

which is fixed-point free on objects and preserves the conference package. On morphisms it retains the same metric normalizer map and relabeling. Since

$$(q_\Pi - 1)(-q_\Pi h) = (q_\Pi - 1)h,$$

its two orbits are exactly the fibres of the forgetful functor. Thus the notation $\mathscr{C}_{\mathrm{ch}}/\langle uq\rangle$ in the theorem means this action groupoid, including its morphisms and isotropy, rather than only an orbit set.

5.1 The dependency lattice

The marking structure is not the Boolean lattice on a list of names. Its basic implications are

$$h \Rightarrow L = Kh, \qquad (q_\Pi, h) \Rightarrow c, \qquad c \Rightarrow \mathrm{Sing}(c) = \Omega_0.$$

Before scalar normalization, forgetting a generator while retaining its line has fibre $K^\times$. After $\alpha^2 = 8^2$, it has the two sheet signs. With $u = -I$, the three nontrivial actions are

$$\begin{aligned} u &: (L, h, c) \longmapsto (L, -h, -c), \\ q &: (L, h, c) \longmapsto (qL, qh, -c), \\ uq &: (L, h, c) \longmapsto (qL, -qh, c). \end{aligned}$$

Here u is an automorphism only after the actual sign has been forgotten, and q is a morphism only in the metric normalizer groupoid, where its outer relabeling σ_τ is part of the morphism. At the fully forgotten vertex, and only there, these transformations give a free transitive Klein-four action.

retained datum	forgetful fibre	sharpness action
line L, no scalar normalization	$K^\times$	scalar trivializations of L
line L, normalized generator	C_2	$u : (h, c) \mapsto (-h, -c)$
oriented c, no selected L	C_2	uq
unordered chordal and conference pairs	$C_2 \times C_2$	u, q

After L is fixed, sheet sign and conference orientation are not independent: $8^{-1}(q_\Pi - 1)$ identifies them. The table therefore records a dependency lattice rather than four unrelated counts.

6 The source functors and their return

We next state exactly what is returned to the earlier papers. Paper II's bridge includes the normalized intertwiner T of Section 2; its pivot 3^{-1} is transported, not recomputed after relabeling. Papers I and III begin with a conference signing and add the choice of L. Paper III's chart lift, scale, outer/Petersen labels, and cross-identification remain bridge data rather than reconstructed outputs.

Corollary 6.1 (Source-by-source marked return)**.** *Relative to the markings in the following table, each of Papers I–III maps to the normalized metric carrier and is recovered at exactly the stated output level.*

paper	*retained source output*	*target decoration*	*inverse returns*
I	*conference switching class, axis carrier, orientation, and selected chordal line L*	*the metric core*	*the same class, carrier, orientation, and L, modulo switching and relabeling*
II	*five-isotypic projected tensor with chosen sheet sign, marked axes, and T*	*W_5, T, and the fixed pivot*	*the same projected tensor, sheet sign, six-axis carrier, and T*
III	*relative conference source, selected L, and full bridge datum*	$\mathfrak{m}_{\mathrm{III}}$	*the same retained source, orientation, L, and bridge datum*

No inverse is asserted outside the declared retained output.

Proof. Paper I's theorem *Support cubic and golden continuation* and its corollary *Nodes, symmetry, and integral commutant* recover the conference signing, orientation, and axis carrier [12]. Paper III's proposition *Relative marked orientation bridge* supplies the same conference source relative to its fixed bridge datum [13]. Given L, Corollary 4.3 recovers the unique normalized $h \in L$. Returning forgets this derived generator and leaves the source data unchanged.

For Paper II, *Balanced sheets and cubic-first orientation* supplies the signed generator. Proposition 3.2 recovers the axis carrier, and Corollary 4.3 supplies c. Writing $B_0 = I + J$, the exact forward and reverse tensor maps are

$$h = 3^{-1}\,\mathrm{Pol}\Big(\mathrm{Sym}^3(B_0 T^{-1})\nu\Big), \qquad \nu = \mathrm{Sym}^3(T B_0^{-1})\,\mathrm{Pol}^{-1}(3h).$$

Thus the pivot, polynomialization, self-duality, and bridge are undone in the reverse order. Every operation is induced by the marked axis relabeling, so the return is natural. □

Corollary 6.2 (Neutral base change)**.** *Every construction and equivalence above commutes with scalar extension K/k. In particular,*

$$Z_{5,K} = \coprod_{P \in \mathrm{Syl}_5(G)} R_K^P \longrightarrow \mathrm{Syl}_5(G)_K$$

is the constant split double cover $(G/C_5)_K \to (G/D_{10})_K$.

Proof. The Reynolds idempotent commutes with scalar extension, the Hankel Jacobian identity is scheme-theoretic over k, and the C_5-fixed eigenlines are already split over k. The conference pair, the normalizer action, and the forward and inverse tensor maps are defined over k; the scalars 3 and 8 remain units after extension. $\square$

This is a theorem about neutral scalar extensions of the fixed carrier. It does not classify arbitrary twisted K-forms and does not create a new matching problem on $\mathbf{P}^1(\mathbf{F}_{11^m})$.

7 The two integral lattices of the six-set

Let Ω be any six-set and put $P = \mathbf{Z}^\Omega$. The reconstruction produces two different integral lattices:

$$A_{\mathrm{aug}}(\Omega) = \ker(P \xrightarrow{\sum} \mathbf{Z}), \qquad D(\Omega) = \{x \in P : \textstyle\sum_\omega x_\omega \equiv 0 \pmod 2\}.$$

The first has rank five and the second has rank six. For the standard permutation pairing,

$$D(\Omega)^\vee = P + \mathbf{Z}\frac{\mathbf{1}_\Omega}{2}.$$

Conflating the two lattices hides both the rank and the factor of two.

Proposition 7.1 (The common six-point heart)**.** *The two lattices canonically determine the same four-dimensional $\mathbf{F}_2S(\Omega)$-module*

$$H_\Omega = \mathrm{Aug}(\mathbf{F}_2^\Omega)/\langle \mathbf{1}_\Omega \rangle.$$

More precisely, reduction of $A_{\mathrm{aug}}(\Omega)$ is $\mathrm{Aug}(\mathbf{F}_2^\Omega)$, while

$$D(\Omega)/2P \cong \mathrm{Aug}(\mathbf{F}_2^\Omega);$$

quotienting the common constant line gives H_Ω.

Proof. A binary vector of even coordinate sum lifts to an integral vector of sum zero by changing one coordinate by an even integer. This proves the first surjectivity. The definition of $D(\Omega)$ proves the second identification. Since $|\Omega| = 6$ is even, the constant binary vector lies in both reductions. All maps are induced by the permutation lattice and are therefore functorial in Ω. $\square$

The rank-five and rank-six outputs meet at H_Ω; they do not become the same lattice.

8 Conference saturation

We now prove the uniform lattice statement behind the golden specialization. The closest classical analogue is Chapman's construction for skew conference matrices and imaginary quadratic orders [3]; the symmetric spectral setting is discussed by Haemers–Parsaei Majd [8]. We need only the elementary real, root–weight statement below.

Theorem 8.1 (Symmetric-conference root–weight saturation)**.** *Let B be a normalized symmetric conference matrix of order $n \equiv 2 \pmod 4$, so $B^2 = (n-1)I$, and put*

$$L = \mathbf{Z}^n, \qquad h = \frac{\mathbf{1}}{2}, \qquad \varphi_B = \frac{I + B}{2}.$$

Then $D_n^\vee = L + \mathbf{Z}h$ *is the minimal over-lattice of* L *stable under* φ_B*, independently of switching, and*

$$\varphi_B^2 - \varphi_B = \frac{n-2}{4} I.$$

The coefficient algebra acting faithfully on $D_n^\vee / 2D_n^\vee$ *is*

$$\mathbf{F}_2[t] / \left(t^2 + t + \frac{n-2}{4} \right).$$

It is $\mathbf{F}_4$ *for* $n \equiv 6 \pmod 8$ *and* $\mathbf{F}_2 \times \mathbf{F}_2$ *for* $n \equiv 2 \pmod 8$*.*

Proof. Normalize the first row of B to be positive off the diagonal. Every column of $I + B$ is odd, so $\varphi_B L \subset L + \mathbf{Z}h$, while

$$\varphi_B e_0 = h.$$

Thus every stable over-lattice containing L contains h. Orthogonality with the first row gives row sums $n-1, 1, \ldots, 1$, and therefore, with $m = (n-2)/4$,

$$\varphi_B h = h + m e_0.$$

This proves stability and minimality. A diagonal switching matrix sends h to h modulo L, so the lattice is switching-independent.

The polynomial identity follows directly from $B^2 = (n-1)I$. Reducing it modulo two gives the displayed algebra. When m is odd the polynomial is $t^2 + t + 1$, hence the algebra is $\mathbf{F}_4$. When m is even it is $t(t+1)$. The action is not a scalar quotient: in the basis $h, e_1, \ldots, e_{n-1}$,

$$\bar{e}_0 = \sum_{i>0} \bar{e}_i, \qquad \bar{\varphi}_B(\bar{e}_0) = \bar{h},$$

so $\bar{\varphi}_B$ is neither zero nor the identity. The split algebra therefore acts faithfully. □

9 The golden residue

Specialize Theorem 8.1 to the oriented order-six conference operator recovered above. Put

$$L^{\max} = D_6^\vee, \qquad \varphi = \frac{I+B}{2}, \qquad M = L^{\max}/2L^{\max}.$$

Then $\varphi^2 - \varphi - 1 = 0$, so M is a three-dimensional vector space over $\mathbf{F}_4 = \mathbf{F}_2[\bar{\varphi}]$. This normalization-before-reduction is essential:

$$\mathbf{Z}[B]/2 \simeq \mathbf{F}_2[\epsilon]/(\epsilon^2), \qquad \mathbf{Z}[\varphi]/2 \simeq \mathbf{F}_4.$$

Proposition 9.1 (The commutator heart)**.** *The submodule* $H = [G, M]$ *is the unique nonzero proper* $\mathbf{F}_4 G$*-submodule of* M*. It has dimension two over* $\mathbf{F}_4$*,*

$$\ell := M/H \cong \mathbf{F}_4$$

with trivial G*-action, where the quotient line* ℓ *is canonical but its displayed trivialization is not. The sequence*

$$0 \longrightarrow H \longrightarrow M \longrightarrow \ell \longrightarrow 0$$

is nonsplit. As an $\mathbf{F}_2 G$*-module,* $H \cong H_{\Omega_0}$*.*

Proof. Use the integral basis $h, e_1, \dots, e_5$ of $D_6^\vee$. Directly from the signed permutation action,

$$H = \left\{ (0, x_1, \dots, x_5) : \sum_{i=1}^{5} x_i = 0 \right\}.$$

The oriented stabilizer commutes with B, hence with φ, so $[G, M]$ is stable under $\bar{\varphi}$. It therefore has dimension two over $\mathbf{F}_4$. The quotient is an $\mathbf{F}_4$-line, and G acts trivially because G is perfect. Here is the full nonsplitting calculation. Take the generators whose underlying six-point permutations are

$$(0, 1, 3, 2, 5, 4), \qquad (1, 2, 0, 5, 3, 4);$$

they have orders two and three and generate the transitive A_5. Their actions on a vector $a = (a_0, a_1, \dots, a_5) \in M$, written in this binary basis, are

$$\begin{aligned} s(a) &= (a_0, a_1, a_3, a_2, a_5, a_4), \\ t(a) &= (a_0, a_2, a_1 + a_2, a_2 + a_4, a_0 + a_2 + a_5, a_0 + a_2 + a_3). \end{aligned}$$

If both fix a, then the last four coordinates in the first equation and the first three nontrivial equations in the second give successively

$$a_2 = a_3, \qquad a_4 = a_5, \qquad a_1 = a_2 = 0, \qquad a_3 = a_4 = 0;$$

the remaining equation gives $a_0 = 0$. Thus $M^G = 0$. Since G is perfect, every one-dimensional representation over $\mathbf{F}_4$ is trivial. Therefore M has no invariant $\mathbf{F}_4$-line, and the sequence is nonsplit.

For completeness, the commutators of the same two generators span the four binary vectors

$$\begin{aligned} &(0, 1, 1, 0, 0, 0), \quad (0, 0, 0, 1, 1, 0), \\ &(0, 1, 0, 1, 0, 0), \quad (0, 0, 1, 0, 0, 1), \end{aligned}$$

which proves the displayed formula for $H = [G, M]$. A proper nonzero $\mathbf{F}_4 G$-submodule of the two-dimensional space H would be an invariant line, hence cannot exist. Any second invariant plane in M would meet H in such a line. Therefore H is the unique nonzero proper submodule.

We also record the representation-theoretic identification needed below. The action of G on H is nontrivial, hence faithful because A_5 is simple. Its determinant is trivial because A_5 is perfect. Therefore its image lies in $\mathrm{SL}_2(\mathbf{F}_4)$; both groups have order 60, so the action identifies G with $\mathrm{SL}_2(\mathbf{F}_4)$ and H with one of its two Frobenius-conjugate natural modules. These two modules are nonisomorphic: on a fixed class of elements of order five their traces are respectively ω and ω^2. Schur's lemma, followed by restriction of scalars, gives

$$\mathrm{End}_{\mathbf{F}_2 G}(H) = \mathbf{F}_4.$$

Indeed, an additional Frobenius-semilinear endomorphism would intertwine the two nonisomorphic natural modules.

Finally, the inclusion $\mathbf{Z}^{\Omega_0} \subset D_6^\vee$ induces a G-map

$$\mathrm{Aug}(\mathbf{F}_2^{\Omega_0}) / \langle \mathbf{1} \rangle \longrightarrow M.$$

Indeed,

$$\mathbf{Z}^{\Omega_0} \cap 2D_6^\vee = 2\mathbf{Z}^{\Omega_0} + \mathbf{Z}\mathbf{1}.$$

The image of the augmentation subspace is exactly the displayed formula for H. This gives the claimed canonical isomorphism with H_{Ω_0}. $\square$

Proposition 9.2 (The extension line)**.** *For either Frobenius-conjugate natural module H of $A_5 \cong \mathrm{SL}_2(\mathbf{F}_4)$,*

$$\mathrm{Ext}^1_{\mathbf{F}_4 A_5}(\mathbf{F}_4, H) \cong \mathbf{F}_4.$$

All three nonzero classes have isomorphic middle modules, and $D_6^\vee/2D_6^\vee$ is that unique nonsplit middle module.

Proof. Take

$$s = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad t = \begin{pmatrix} 0 & \omega \\ \omega^2 & 1 \end{pmatrix}$$

in $\mathrm{SL}_2(\mathbf{F}_4)$. Their orders are $2, 3$, and the order of st is 5. A cocycle is determined by $a = f(s)$ and $b = f(t)$. The triangle relations impose

$$\begin{aligned} (1+s)a &= 0, \\ (1+t+t^2)b &= 0, \\ (1 + st + \cdots + (st)^4)(a + sb) &= 0. \end{aligned}$$

On H, the three relation norms have ranks $1, 0, 0$: the first is the rank-one transvection norm, $t^2 + t + 1 = 0$, and $st - 1$ is invertible while $(st)^5 = 1$. Hence $\dim_{\mathbf{F}_4} Z^1 = 1 + 2 = 3$. The coboundary map

$$H \longrightarrow H \oplus H, \qquad v \longmapsto ((s-1)v, (t-1)v)$$

is injective because $H^G = 0$, so $\dim_{\mathbf{F}_4} B^1 = 2$. This proves the Ext formula.

Scalar automorphisms of the two endpoints act transitively on the three nonzero vectors of the Ext line. They therefore have isomorphic middle modules. Proposition 9.1 shows that M is nonsplit, so it realizes the unique nonsplit isomorphism class. This agrees with the characteristic-two block description recorded by Bleher [2, §3.3]. □

Theorem 9.3 (Golden orientation becomes Frobenius)**.** *Restriction of $\bar{\varphi}$ to H is one of the two primitive elements ω, ω^2 of*

$$\mathrm{End}_{\mathbf{F}_2 G}(H) = \mathbf{F}_4.$$

Reversing the conference orientation and applying the outer normalizer both exchange these elements by Frobenius. Consequently there is a Cartesian square of principal C_2-torsors

$$\begin{tikzcd} \{[B], [-B]\} \arrow[r] \arrow[d] & \{\omega, \omega^2\} \arrow[d] \\ \left\{ \substack{\textit{unoriented} \\ \textit{conference class}} \right\} \arrow[r] & \left\{ \substack{\textit{unordered} \\ \textit{exotic class}} \right\} \end{tikzcd}$$

relative to the recovered six-set.

Proof. The equation $\bar{\varphi}^2 + \bar{\varphi} + 1 = 0$ makes its restriction to the simple heart primitive. Reversal sends

$$B \longmapsto -B, \qquad \varphi \longmapsto 1 - \varphi,$$

which reduces to $\omega \mapsto 1 + \omega = \omega^2$. The outer normalizer exchanges the two G-invariant conference orientations and therefore acts by the same involution. Both vertical maps are principal C_2-torsors, and the top map is equivariant and bijective, so the square is Cartesian. □

The lattice theorem and the modular extension theorem are classical in nearby but separate forms: Chapman treats the skew/imaginary conference analogue, while Bleher records the relevant uniserial block modules. The role of Theorem 9.3 is their composition with the orientation reconstructed in this paper.

10 Residual Frobenius markings

Paper IV ends with a different residual marking. The comparison is governed by the following standard descent lemma, not by a geometric arrow between the papers.

Lemma 10.1 (Frobenius-orbit commutant). *Let V be a simple $\mathbf{F}_qG$-module. Suppose that after scalar extension to an algebraic closure it is a multiplicity-free orbit*

$$V_0 \oplus V_0^{(q)} \oplus \cdots \oplus V_0^{(q^{r-1})}$$

of r absolutely simple pairwise nonisomorphic constituents. Then $\mathrm{End}_{\mathbf{F}_qG}(V) \cong \mathbf{F}_{q^r}$. *Selecting one constituent, equivalently one embedding of this commutant field, is a principal* $\mathrm{Gal}(\mathbf{F}_{q^r}/\mathbf{F}_q)$*-rigidification.*

Proof. After scalar extension, Schur's lemma identifies the commutant with the product of r scalar copies of the algebraic closure. Frobenius permutes the factors transitively. Its fixed algebra is $\mathbf{F}_{q^r}$, and its embeddings are permuted simply transitively by the Galois group. □

For the upper branch, Proposition 9.1 proves that H_{Ω_0} has commutant $\mathbf{F}_4$; its two constituents and the two primitive scalars form the C_2-torsor of Theorem 9.3. Paper IV's result *The operator field* proves that its binary code is simple, becomes a multiplicity-free Frobenius orbit of three twelve-dimensional constituents, and has commutant $\mathbf{F}_8$. Its recovered operators are

$$\{\alpha, \alpha^2, \alpha^4\},$$

a principal C_3-rigidification [14]. The two examples share the residual-field mechanism. They have different groups, modules, bases, and geometries, and no map between their carriers is asserted.

This completes the proof of Theorem 1.3 and explains the last arrow in the series map: Paper V does not merge the upper and lower branches; it identifies the common reason that their final ambiguity is a Frobenius orbit.

11 Verification and trust boundary

Every load-bearing argument above is printed in the manuscript: the Hankel Jacobian saturation, the two-point chordal classification, the stabilizer quotient, the outer-difference bridge, the groupoid fibres, the root–weight saturation, the natural-module commutant, and the triangle-presentation Ext calculation. The paper-owned standard-library checker

`verification/evidence/paper_ii_chordal_axis.py`

has schema `paper-v-chordal-axis-v2`. It independently reconstructs the matching quotient from Paper II's frozen arithmetic, computes the central projector and both intertwiner spaces, checks all cubic coefficients, enumerates the rational singular loci, and determines the pencil action. Its exact output is the adjacent JSON file. The JSON names, sizes, and hashes both frozen inputs. Their SHA-256 values begin `d2100485` for the matching-module source and `d45512c9` for the orbit-scout data; the unabbreviated values are part of the machine-readable proof interface. From the paper directory the paper-local replay is

```
make evidence
```

and returns `CHECK OK (NO_MATCH)`; `NO_MATCH` records the failure of the literal conference-line identification, not a failed check. A separately written cold replay reproduced the normalized coefficient vector; its full byte SHA-256 is recorded in the adjacent JSON certificate. The execution is replay evidence for Proposition 2.1, not a premise of the classification theorem. In particular, it substitutes for none of Lemma 3.1, Proposition 3.2, Lemma 3.4, Theorem 8.1, or Proposition 9.2.

12 Conclusion

The chordal and conference cubics are geometrically distinct, but after a chordal line is selected they recover the same marked six-axis carrier. The singular quartic of the chordal cubic recovers the constant double cover $A_5/C_5 \to A_5/D_{10}$, and the normalized outer difference gives mutually inverse reconstruction between a normalized chordal generator and an oriented conference generator. Forgetting the selected chordal line is exactly the free C_2-quotient. Thus the result identifies the source information retained by the two cubic shadows without identifying the two invariant cubic lines themselves.

The recovered six-set also controls the integral normalization. The rank-five augmentation and rank-six root–weight lattices remain distinct, while their common binary heart carries the $\mathbf{F}_4$-structure on which conference reversal is Frobenius. Together these results determine both the common marked carrier and the precise residual ambiguity left by the reconstruction.

Clebsch: Rigidity from Sparse Shadows

I From deep holes, a form arises;
II beneath it, paired chords turn true;
III through many veils, the golden thread runs;
IV from bare, whispered words, a plane rises;
V **beneath one form, two shadows, a hidden twist between.**

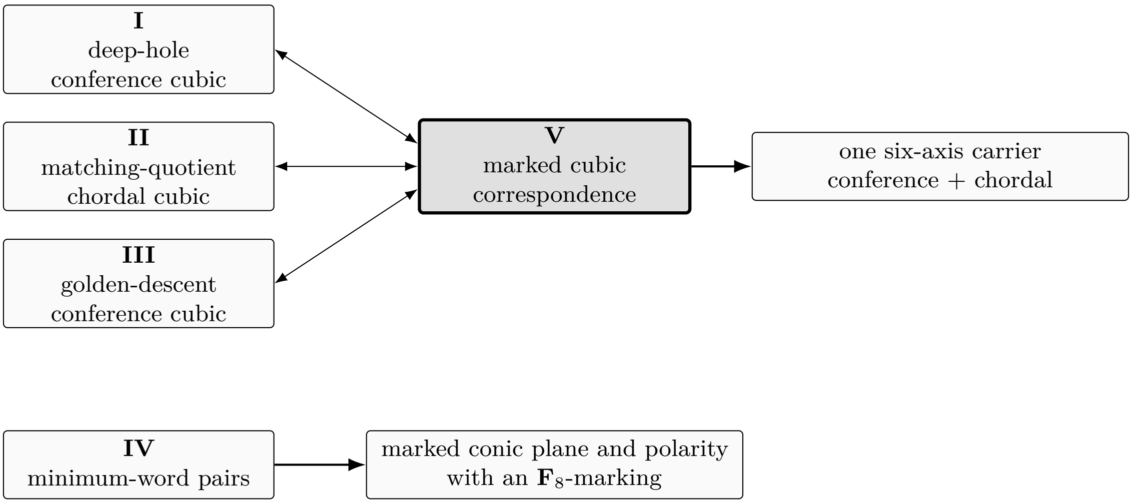


Figure 1: Programme map. The upper double arrows are Paper V's marked transports and returns between the chordal and conference shadows; the right arrow is its common-carrier reconstruction. The lower solid arrow is Paper IV's independent minimum-word reconstruction. Only the residual C_2- and Frobenius-orbit markings are compared across the two branches.

The five papers are logically independent; the map records reconstruction correspondences and thematic relations, not proof dependencies.

The scattered shadows gather on one carrier, but not by becoming equal. Papers I and III reach the conference shadow from deep-hole and golden-descent data; Paper II reaches the chordal shadow from matching-quotient data. Paper V shows that these distinct cubic shadows recover the same marked six-axis carrier, with a residual C_2-torsor when the selected chordal line is forgotten.

Paper IV forms an independent branch. There the sparse data are minimum-word pairs rather than cubic data, and weighted pair concurrences recover a marked conic plane and polarity. Its residual marking is governed instead by Frobenius on the $\mathbf{F}_8$-commutant. The two branches therefore illustrate the same information-loss principle without sharing a geometric carrier: reconstruction can recover the source while leaving a small, rigid residual marking.

Distinct shadows need not become isomorphic in order to encode equivalent source data; what matters is whether the ambiguity left after reconstruction can itself be isolated and rigidified.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted with proof exploration, literature research, exact computation, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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